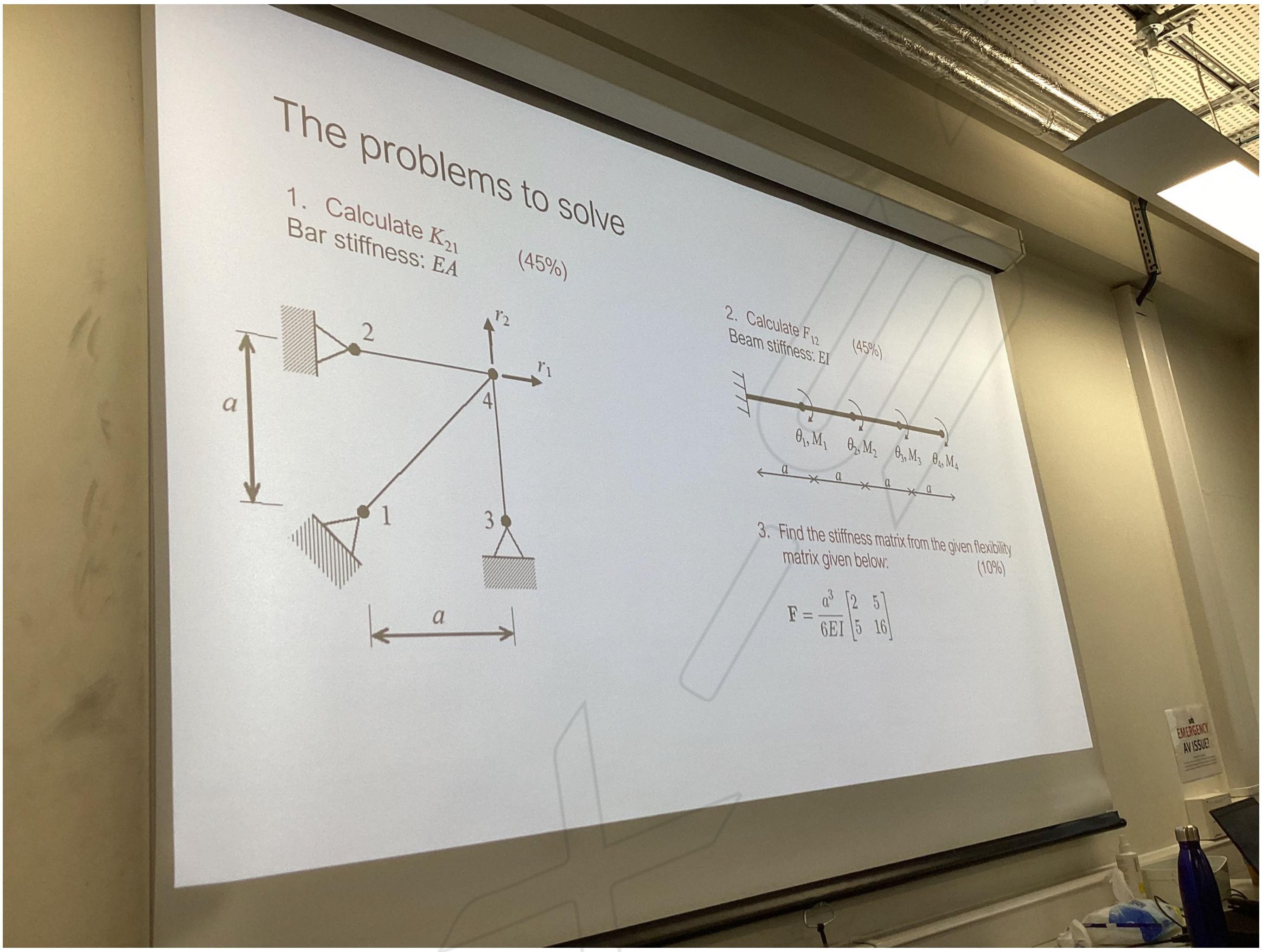
10.26



1. Let $r_1 = 1,\ r_2 = 0$

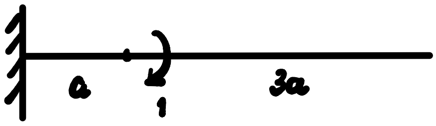
$\therefore\ T_{34} = 0$

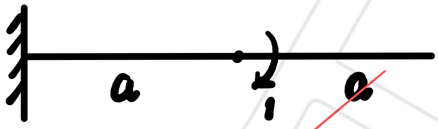
$T_{14} = \frac{\Delta L}{L} EA = \frac{\frac{\sqrt{2}}{2}}{\sqrt{2}a} EA = \frac{1}{2}\frac{EA}{a}$

$T_{24} = \frac{EA}{a}$

$\because\ \Sigma F_V = R_2 - T_{34} - \frac{\sqrt{2}}{2} T_{14} = 0$ $\therefore\ R_2 = \frac{\sqrt{2}}{4}\frac{EA}{a}$

$\therefore\ K_{12} = R_2 = \frac{\sqrt{2}}{4}\frac{EA}{a}$

2. 



$$\therefore \quad M_1 = \begin{cases} 1 & (0 \le z \le a) \\ 0 & (a < z \le 4a) \end{cases}$$

$$\therefore \quad M_2 = \begin{cases} 1 & (0 \le z \le 2a) \\ 0 & (2a < z \le 4a) \end{cases}$$

$$\therefore \quad F_{12} = \int \frac{M_1 \bar{M}_2}{EI} dz = \int_0^a \frac{1}{EI} dz = \frac{a}{EI}$$

3. $K = F^{-1}$

$$= \frac{adj(F)}{det(F)}$$

$$= \frac{C^T}{det(F)}$$

$$= \frac{a^3}{6EI} \frac{\begin{bmatrix} 16 & -5 \\ -5 & 2 \end{bmatrix}}{2 \times 16 - 5 \times 5}$$

$$= \frac{a^3}{6EI} \begin{bmatrix} \frac{16}{7} & -\frac{5}{7} \\ -\frac{5}{7} & \frac{2}{7} \end{bmatrix}$$

$$= \frac{a^3}{42EI} \begin{bmatrix} 16 & -5 \\ -5 & 2 \end{bmatrix}$$